

Anteater Poker User Manual

Version 1,0

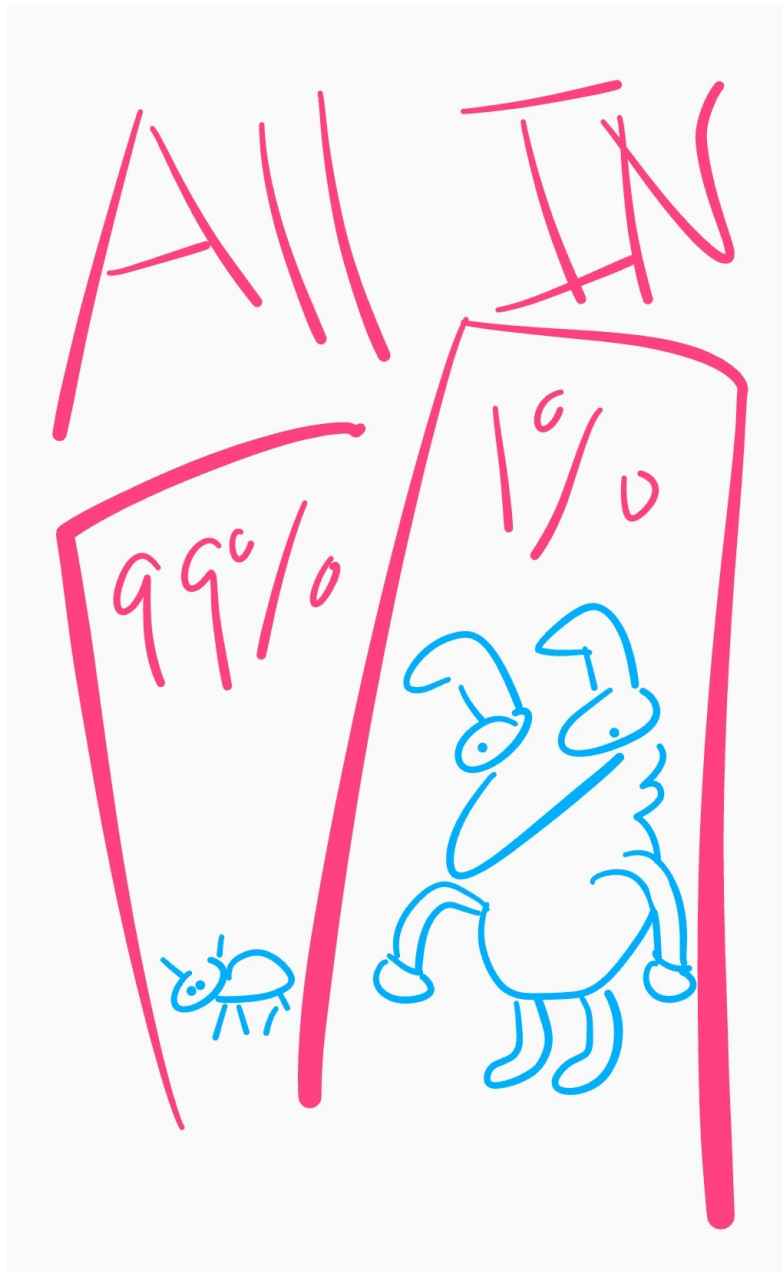
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Affiliation: University of California, Irvine

Team Name: 99%

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Glossary

Term	Definition
All in	When a player bets their remaining chips.
Big Blind	A mandatory bet placed by the player 2 positions to the left of the dealer before any cards are dealt.
Call	Match the current highest bet to stay in the round.
Check	If no bet has been made, pass the turn to the next active player without betting.
Community	The 5 cards that are shared by every player. These cards are revealed in stages.
Dealer	The person that typically deals the cards and shuffles in a typical poker game.
Flop	The stage where there is a betting round after the first 3 community cards are revealed.
Fold	Throw in cards and give up your hand for the current game, forfeiting any chips that are already bet.
Hole	Two private cards that only the player can see, which will be revealed after betting.
Pot	The total amount of chips bet by all players in the current hand.
Raise	Increase the current bet by a minimum raise amount.
River	The stage where there is a betting round after the 5th community card is revealed.
Preflop	The stage where there is an initial betting round where no community cards are revealed.
Showdown	Where every card is revealed and the winner is chosen out of the remaining players.
Small Blind	A mandatory bet placed by the player 1 position to the left of the dealer before any cards are dealt.
Turn	The stage where there is a betting round after the 4th community card is revealed.

1. Online Poker

1.1 Usage Scenario

1. A session begins with the user launching the program
2. Input name and choose card set
3. Connect to server and start playing

Log in Screen

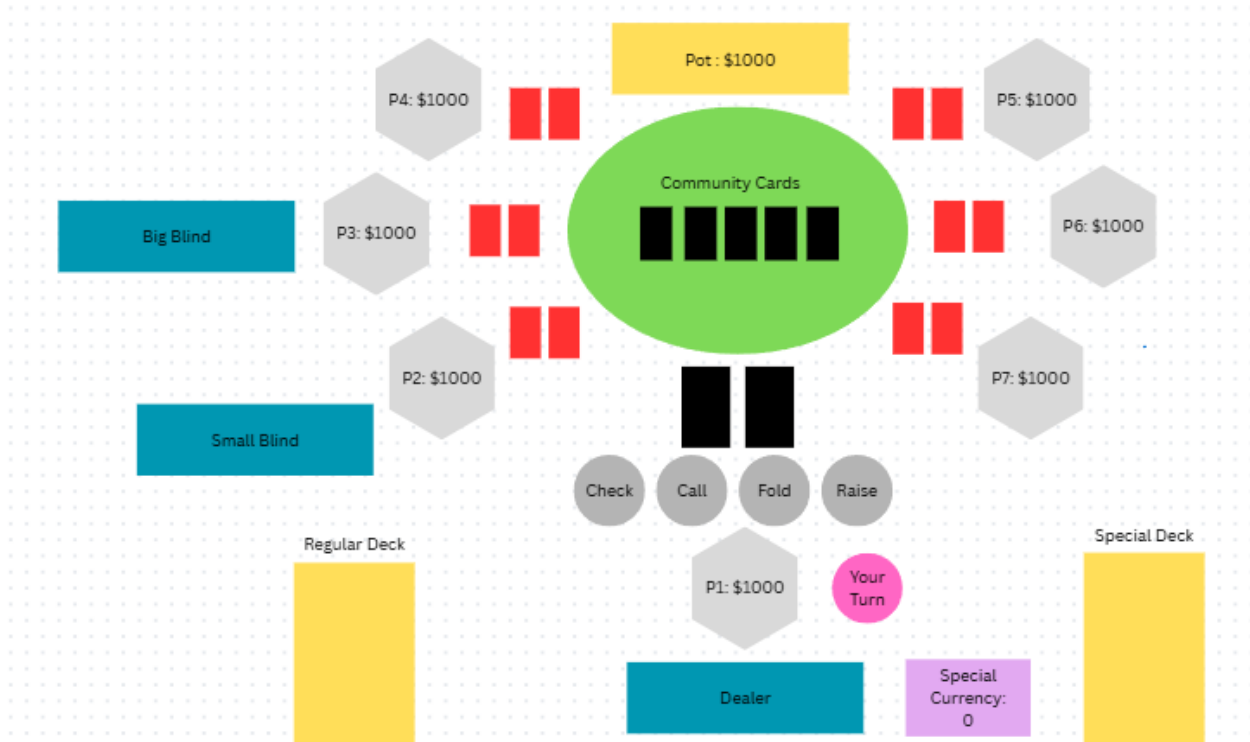
ANTEATER POKER

Insert Name:

Choose Card Set:

CONNECT TO SERVER

Screenshot of Game



1.2 Goals

1. Implement base game of Texas Hold-em Poker including hand hierarchy
 - Standard deck of 52 cards
 - Standard winning poker hands to determine winner of each round
 - Pot distribution depends on winner and ties if applicable
 - Poker game will support multiple players in a single game, including a Poker bot if not enough active players are present

2. Ability to act as a player through raising, checking, calling, or folding
 - Prevent player from performing moves when it is not their turn
 - Prevent illegal moves, such as placing an insufficient bet
 - Bot will also act as a player with the same moveset, but with decision making based on statistics of the current cards at play
3. Utilize special cards and currency to add another layer to regular Texas Hold-em Poker
 - Secondary deck of special cards with special moves and features, including:
 - Swap one card with another player at the table
 - Swap both cards with another player (both must be from the same player)
 - Redraw one card from the main deck
 - Reveal next card on the community cards (flop, turn, or river)
 - See one of your opponent's cards in exchange for showing one of your cards
 - Swap your opponents' cards (requires 3+ players)
 - Secondary currency
 - Amount is dependent of percentage of points lost (ex: loss streak results in more of second currency rewarded)
 - Used to draw from secondary deck of special cards

1.3 Features

1. Texas Hold em Poker with a twist!
2. Server client interaction
 - Server has all the information
 - Client interacts with server
3. Money connected to each player
4. Interactive player GUI
5. Special deck with cards that change how players interact with each other and the game
 - Earned through betting a certain percentage of your current money
 - Cards allow you to:
 - Swap 1 card with someone at the table
 - Swap both cards with someone at the table
 - Redraw one of your cards from main deck
 - Reveal next card on the river
 - See one of your opponent's cards in exchange for showing one of yours
 - Swap your opponents cards (3+ players)
6. Secondary currency to access special bonuses
 - Based on how much percentage you lose of your money
 - You get more if you're on a loss streak
 - Use to draw from extra deck/choose from specific deck
 - Pay to add certain chance to get better card

2. Installation

2.1 System requirements

Operating System: RHEL8 64-bit Linux

CPU: 64-bit architecture

Ram: 1.5 GB of RAM

Storage: 10 GB

Software: GTK (version 3.0)

2.2 Setup and configuration

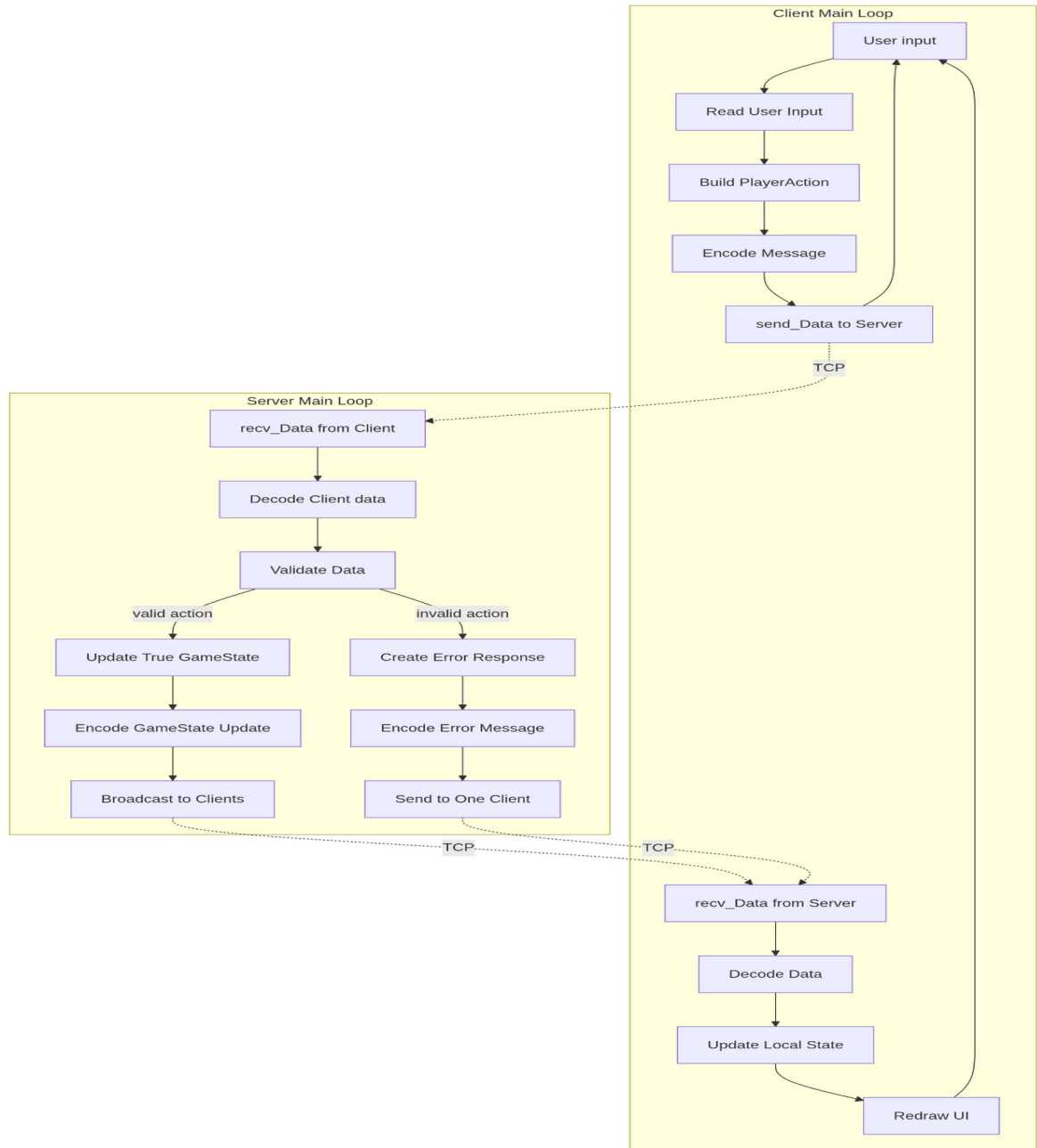
1. Download the user package from the server
(team16@bondi.eecs.uci.edu/repos/poker.git)
 2. Extract the user package using the following commands:
 - % `gtar xvzf BinaryArchive.tar.gz`
 - % `evince poker/doc/Poker_UserManual.pdf`
 - % `poker/bin/server &`
 - % `poker/bin/poker`
- The above commands allow the user to extract the poker folder using [gtar] and view the documentation using the document viewer with [evince]. It will also execute the online poker program.

2.3 Uninstalling

1. Execute the following commands:
 - % `rm -rf poker`
- The above commands allow the user to remove the extracted poker file, and the user package.

3. Poker Program Functions and Features

3.1 Client and Server Communication



3.2 Dealer Choice and Card Distribution

1. The initial dealer will be the first player to join the game.
2. The dealer's role gets passed on to the next player in a clockwise direction.

3. The server is able to shuffle the deck before dealing 2 hole cards to each player.
4. The server also places 5 community cards:
 - a. The first 3 cards will be revealed in the Flop stage
 - b. The next card is revealed in the Turn stage
 - c. The final card is revealed in the River stage

User Input: Button (Shuffle)

Program Output: Cards Dealt, Cards Revealed after Each Round

Logging into the game

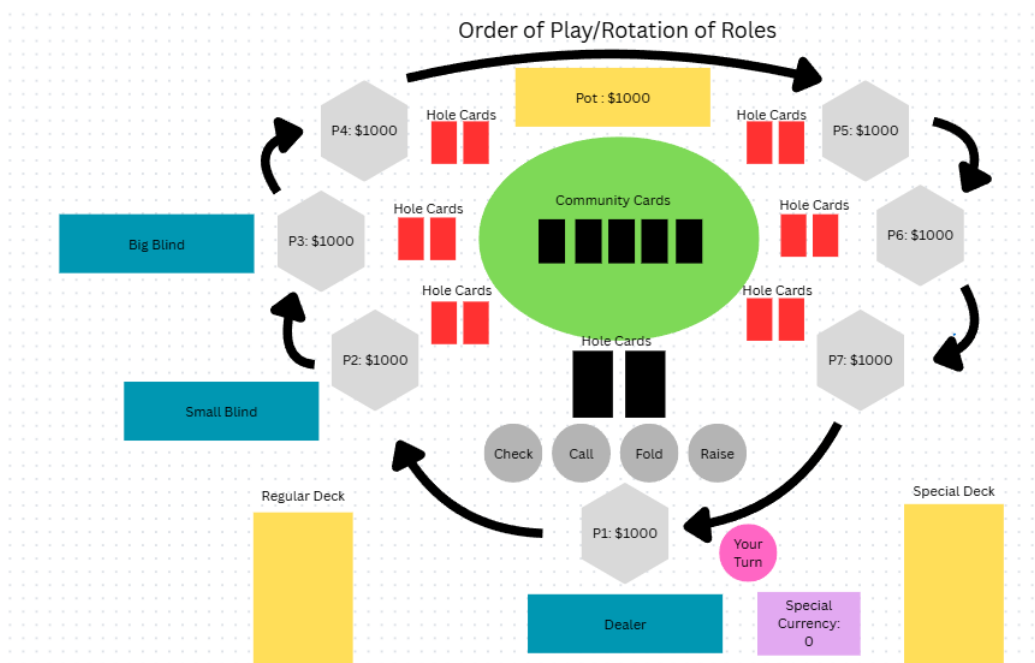
ANTEATER POKER

Insert Name:

Choose Card Set:

CONNECT TO SERVER

Diagram of Game Layout



3.3 Poker Game Integration

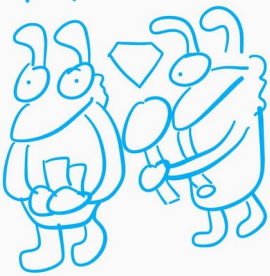
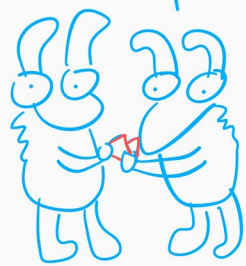
1. The server shuffles the deck and deals 2 hole cards to each player.
2. Betting begins on the active player that is left of the dealer and proceeds clockwise.
3. On each turn, a player may do the following: Fold, Check, Call, Raise
4. The game will proceed through stages:
 - a. Preflop: first betting round
 - b. Flop: betting round after server reveals first 3 community cards
 - c. Turn: betting round after server reveals 4th community card
 - d. River: betting round after server reveals 5th and final community card***The betting rounds continue until all players match the current bet.***
5. After the River stage, all remaining players reveal their 2 hole cards.
6. The server evaluates each player's hand (2 hole cards + 5 community cards).
7. The player with the highest hand wins the pot
If there is a tie, the pot is split evenly amongst the players who tied.
8. The dealer role shifts to the next player clockwise and a new round begins.

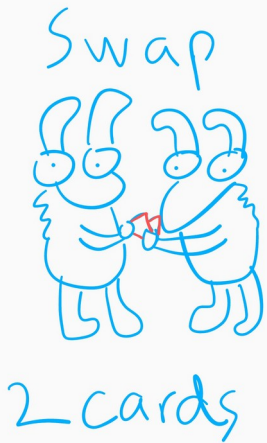
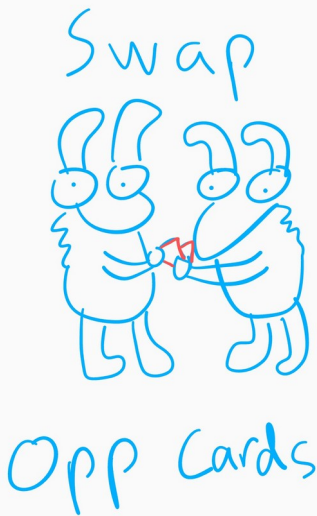
User Input: Button (Fold, Check, Call Raise)
Program Output: Interaction with Server, Updating GUI,

3.4 Special card feature

Special card	function
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Redraw 		Redraw one card from the main deck
Reveal  Community		Reveal next card on the community cards (flop, turn, or river)

Reveal  Opponent		See one of your opponent's cards in exchange for showing one of your cards
Swap  1 card		Swap one card with another player at the table

		Swap both cards with another player (both must be from the same player)
		Swap your opponents' cards (requires 3+ players)

Copyright

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Error Messages

Error	Meaning
ERROR: No port provided	Missing port number
ERROR, no such host	No host exists for the connecting end
ERROR connecting	Failed to connect to host
ETIMEDOUT	Connection establishment timed out without establishing a connection
ENETUNREACH	The network is not reachable from this host.
EISCONN	The socket is already connected.
ENAMETOOLONG	The length of the path argument exceeds {PATH_MAX}.
ERROR writing to socket	Fail to write to the socket
ERROR reading from socket	Fail to read from the socket

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